Major congenital malformations in women with gestational diabetes mellitus: a systematic review and meta-analysis.

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BACKGROUND: The risk of major congenital malformations (MCM) is increased in women with pregestational diabetes mellitus (PGDM). Whether this risk is increased in gestational diabetes mellitus (GDM) is still debated. The aim of this study was to perform a systematic review (and meta-analysis) of major congenital malformations in women with gestational diabetes versus a reference population. METHODS: We conducted a MEDLINE search (1 January 1995 to 31 December 2009) of original studies reporting data on major congenital malformations in women with gestational diabetes and a reference group. Information on pregestational diabetes was collected when available. Two investigators considered studies for inclusion and extracted data; discrepancies were solved by consensus. Meta-analysis tools were used to summarize results. MOOSE and PRISMA guidelines were followed. RESULTS: Two case control and 15 cohort studies were selected out of 3488 retrieved abstracts. A higher risk of major congenital malformations was observed in offspring of women with gestational diabetes with the following relative risk (RR)/odds ratios (OR) and 95% confidence intervals (CI): RR 1.16 (1.07-1.25) in cohort studies and OR 1.4 (1.22-1.62) in case control studies. Risk of major congenital malformations was much higher in offspring of women with PGDM than in those of the reference group: RR 2.66 (2.04-3.47) in cohort studies and OR 4.7 (3.01-6.95) in the single case control study providing information. CONCLUSION: There is a slightly higher risk of major congenital malformations in women with gestational diabetes than in the reference group. The contribution of women with overt hyperglycemia and other factors could not be ascertained. This risk, however, is much lower than in women with pregestational diabetes.